PITCH DECK V2 · 2026 EDITION

AML Open *Framework*

An anti-money-laundering program you can show to your regulator without a six-week reconstruction. Compliance writes it once, engineering runs it as-is, the auditor replays the same answer. No re-translation between teams.

19 SLIDES · 24 CLI COMMANDS · 1,632 TESTS
9 SPECS · 4 JURISDICTIONS · MIT LICENCE

github.com/tomqwu/aml_open_framework

ACT I · THE GAP

Three teams write the program. None of them write the same thing.

1ST LINE · COMPLIANCE

"We wrote it down. In English. In a binder."

What lives here: typology narratives, threshold rationale, regulator mapping. All Word, PowerPoint, SharePoint.

```
Risk_Assessment_2024.docx
Typology_Standards_v18.pptx
SOP_Approval_Email_Chain.eml
→ archived, not executable
```

ENGINEERING

"We shipped detectors. The SQL is in the warehouse."

What lives here: SQL behind a vendor abstraction, Airflow DAGs, model code, Slack-signoff threshold tweaks.

```
rule_engine.dbt/
airflow/dags/aml_pipeline.py
commits.merged_3am.signoff=Slack
→ no link back to the binder
```

2LOD · AUDIT · REGULATOR

"Six weeks to reconstruct what fired, why, and who decided."

What lives here: sample-based testing, e-mail forensics, analyst interviews, redacted screenshots — months after the fact.

```
Audit_Walkthrough_Q3.xlsx
decisions_extracted_manually.csv
sample_size = 25 of ~3,000
→ findings often "cannot evidence"
```

Across recent enforcement orders, regulators rarely allege missed typologies — they allege the bank **cannot evidence** *what it did.*

ACT I · THE GAP

The status quo costs more than you think.

BSA ENFORCEMENT · 2020-2024

$36*B+*

Cumulative AML/BSA fines against US financial institutions. Single-bank actions routinely cross $1B (TD 2024 · $3B; Wells Fargo 2024 · $35M; Danske 2022 · $2B).

SOURCE · FINCEN ENFORCEMENT INDEX

AVERAGE ENFORCEMENT ORDER DURATION

18*mo*

How long an institution operates under a consent order before remediation closes. Most cite the same root cause: the bank could not *evidence* what its controls did.

SOURCE · 2026 PAIN POINTS RESEARCH (PAIN-2)

DAILY LEADER TIME ON RECONCILIATION

22*min*

What a CCO / MLRO spends every weekday cross-referencing three dashboards, one e-mail thread, and Slack escalations to assemble a single picture of yesterday's run.

SOURCE · TIER-1 BANK INTERVIEWS · N=12

Detection is not the bottleneck. **Provability is.**

ACT II · THE SPEC IS THE CONTRACT

One source.
Three readings, all faithful.

CCO · READS TOP-DOWN

Program metadata, regulator, owner. *What is in scope, who is accountable, when it took effect.*

→ board report · regwatch · effectiveness pack

MLRO · READS SIDEWAYS

Rule logic + regulation refs + tuning grids. *Is each rule still earning its keep?*

→ backtest · propose-change · MRM bundle

aml.yaml · canadian_schedule_i_bank

```yaml
# Program — what compliance owns
program:
  name:            schedule_i_bank_aml
  jurisdiction:    CA
  regulator:       FINTRAC
  owner:           chief_compliance_officer
  effective_date:  2026-01-01

# Rules — what engineering ships
rules:
  - id:       structuring_cash_deposits
    severity: high
    regulation_refs:
      - citation: "PCMLTFA s.11.1"
    logic:
      type:     aggregation_window
```

ENGINEERING · READS AS SCHEMA

JSON Schema → Pydantic → DuckDB. *Same spec the engine compiles, same fields the API exposes.*

→ run · validate-data · audit chain

AUDITOR · READS AS EVIDENCE

SHA-256 of this exact file. *Tied to every alert, decision, and STR — replayable byte-for-byte.*

→ replay · auditor-pack · regwatch verify

The spec is the contract.

ACT II · THE SPEC IS THE CONTRACT

Every leader question has a one-line answer.

Compliance writes the spec.
Engineering ships from the spec.
The auditor replays the spec.

Same source, three readings. Anything a board, regulator, or analyst asks for can be answered in < 60 seconds — no vendor support cycle, no analyst pulling all-nighters to reconstruct a quarter.

CCO · MLRO

- `aml today` Morning checklist · per persona
- `aml propose-change` 2LoD sign-off packet
- `aml backtest` Is rule X still earning its keep?
- `aml effectiveness-pack` FinCEN NPRM 2026 evidence

AUDITOR · 2LOD

- `aml auditor-pack` Self-service ZIP · no IT call
- `aml replay` Historical run · byte-for-byte
- `aml mrm-bundle` SR 26-2 model dossier
- `aml regwatch` Reg-text drift detection

ANALYST · INVESTIGATOR

- `aml run` Execute detectors · seeded
- `aml dashboard` 26 pages · persona-aware
- `aml export-amla-str` AMLA RTS draft 2026-02
- `aml pkyc-scan` Periodic KYC trigger sweep

ENGINEERING · SPEC

- `aml init` Scaffold a spec in 60s
- `aml byod` Map warehouse → contracts
- `aml add-rule` New typology · wizard
- `aml typology-import` Curated catalogue · 8 rules

ACT III · THE DAILY RITUAL

Replace four dashboards. The morning, in one view.

01 **Replaces four sources** the leader currently reconciles by hand: Audit dashboard, Investigations queue, escalation email, Slack channel. ~22 minutes every weekday.

02 **Persona-aware.** CCO sees board signal · MLRO sees model-challenge work · analyst sees their queue. Same data, three readings.

03 **Quiet defaults.** When nothing is on fire, the checklist is one line: "All clear". No fake-data dashboards keeping the ritual alive.

Each item points back to another `aml ___` command — the morning view is the framework's navigation.

aml today --persona cco · examples/canadian_schedule_i_bank/aml.yaml

```
$ aml today --persona cco examples/canadian_schedule_i_bank/aml.yaml

# Today · CCO · schedule_i_bank_aml
# Generated 2026-04-29T02:08:34

[1] ℹ Re-run regulation-drift watcher
    FinCEN BOI was silently narrowed in March 2025;
    the watcher catches that pattern. Cron weekly.
    → aml regwatch <spec>

[2] ℹ Verify the hash chain on today's run
    If the chain breaks, every downstream attestation
    fails. The Audit & Evidence page does this in
    one click.
    → aml dashboard <spec>  # → Audit & Evidence → ▯
```

ACT III · THE DAILY RITUAL

"We want what RBC ships." Already in the box.

01 **Curated catalogue.** 8 vetted typologies — TD 2024 findings, FATF 2026 cyber-fraud, UK PSR PS24/3, FinCEN advisories — every rule cites the regulator that justifies it.

02 **Atomic install.** One command splices the rule into your spec, validates against the loader, rolls back if anything breaks. You never see a half-broken YAML.

03 **Or write your own in 60 seconds.** `aml add-rule` handles structuring, velocity, and high-risk-jurisdiction patterns — same audit-citation discipline.

From "we don't know what to start with" to a regulator-cited rule running on real data — under one minute, no vendor meeting required.

aml typology-list · aml typology-import

```
$ aml typology-list --jurisdiction CA

Curated typology catalogue
```

id	sev	source
structuring_cash	high	TD 2024 #2
mule_burst_inbound	high	FinCEN A-2020-A003
wire_to_high_risk_juris	high	FATF R.19
shell_company_pass_through	high	TD 2024 #3
dormant_reactivation	med	TD 2024 #5
ramp_up_then_drain	crit	FATF 2026 #5
pig_butchering_outbound	crit	FATF 2026 #4
app_fraud_first_use_payee	high	UK PSR PS24/3

```
$ aml typology-import structuring_cash.spec.yaml
```

ACT III · THE DAILY RITUAL

Threshold changes leave the email thread. They join the audit trail.

01 **1LoD proposes.** One command produces a 6-section packet — rationale, expected impact, full diff, citations, and a sign-off block.

02 **2LoD reviews in-line.** The packet attaches to the spec PR. Review happens against the diff, in GitHub — not in Outlook, not in Word.

03 **MLRO countersigns.** Signed packet + commit history together are the audit-trail artifact. Eighteen months from now, the answer to "who decided?" is not in someone's inbox.

SR 26-2 / OSFI E-23 / FCA CASS expect the same thing: an evidenced governance ritual. This is what one looks like.

mlro-review-packet.md · attached to spec PR

MLRO Review Packet · `structuring_cash_deposits`

Proposer: Aisha N. (Senior Analyst, FCC) **Generated:** 2026-04-29 02:23 UTC
Spec: `examples/canadian_schedule_i_bank/aml.yaml`

1 · RATIONALE

Lower `sum_amount` threshold from $20k to $15k. Internal precision audit (Q1 2026) flagged 3 suspicious patterns in the 15-20k band that escaped detection. Backtest shows +14% recall, −2% precision.

2 · EXPECTED IMPACT

+12 alerts/month, +1 likely STR.

3 · DIFF

```
having: count: { gte: 3 } - sum_amount: { gte: 20000 } + sum_amount: { gte: 15000 }
```

4 · SIGN-OFF

1LoD PROPOSER Aisha N. 2026-04-29

ACT III · THE DAILY RITUAL

What you'd hand a regulator tomorrow morning.

01 **One ZIP, one command, no IT call.** Hash-verified ledger, examination pack, effectiveness pack, manifest, decisions log — bundled, indexed, auditor-ready.

02 **Self-service.** `--print-link` gives the auditor a deep-link onto the dashboard's Audit & Evidence page in auditor-persona mode. They click before they download.

03 **Hash chain integrity verified at build time.** If the decision log was tampered with — even a single line — the ZIP refuses to build. Tamper-evident, by construction.

Average enforcement order is 18 months because the bank cannot evidence what it did. This is the evidence, already on the shelf.

aml auditor-pack · canadian_schedule_i_bank

```
$ aml auditor-pack examples/canadian_schedule_i_bank/aml.yaml \
   --out auditor-pack.zip --print-link

Auditor pack · schedule_i_bank_aml

┌──────────────────────┬──────────┐
│ Component            │ Status   │
├──────────────────────┼──────────┤
│ MANIFEST.txt         │ ✓        │
│ audit_pack.zip       │ ✓        │
│ manifest.json        │ ✓        │
│ decisions.jsonl      │ ✓        │
│ spec_snapshot.yaml   │ ✓        │
└──────────────────────┴──────────┘

✓ Hash chain verified  Decision log integrity verified
```

ACT IV · THE DETECTION SUBSTRATE

Day-one catalogue. Eight typologies, every line cited.

TYPOLOGY ID	SEVERITY	JURISDICTIONS	SOURCE · CITATION	WHAT IT CATCHES
structuring_cash	HIGH	US · CA · GB · EU	TD 2024 · finding #2 31 CFR 1010.314 · PCMLTFR s.7(1)	≥ 3 cash deposits, total > $25k in 30d, each below the LCTR/CTR floor.
pig_butchering_outbound_fan	CRIT	US · CA · GB · EU · AU · SG	FATF 2026 · finding #4 FinCEN FIN-2023-Alert005	5+ outbound payments to first-use payees in 14d, ≥ $20k total.
app_fraud_first_use_payee	HIGH	GB · EU	UK PSR PS24/3 · Oct 2024 FCA FG24/4	Single push payment ≥ £1k to a payee never paid before.
mule_burst_inbound	HIGH	US · CA · GB · EU	FinCEN FIN-2020-A003 + 2026 RTP/FedNow telemetry	≥ 8 inbound from distinct remitters in 7d, > $15k total.
ramp_up_then_drain	CRIT	US · CA · GB · EU · AU · SG	FATF 2026 · finding #5 FinCEN FIN-2023-Alert005	Dormant account → ≥ $20k inbound → drained ≥ 90% in 72h.
wire_to_high_risk_juris	HIGH	US · CA · GB · EU · AU · SG · HK · JP	FATF R.19 · 31 CFR 1010.610 FinCEN AML/CFT National Priorities	Outbound wire ≥ $3k to a country on FATF grey/black list.
dormant_reactivation	MED	US · CA · GB · EU	TD 2024 · finding #5 FFIEC BSA/AML Manual	Dormant 180+ days → suddenly > $10k traffic in 7d.
shell_company_pass_through	HIGH	US · CA · GB · EU · SG · HK	TD 2024 · finding #3 FATF R.24 · FinCEN BOI rule	Business: ≥ $50k in, > 90% out same day, EoD bal < 5%.

ACT IV · THE DETECTION SUBSTRATE

MLRO challenges the model. Without a six-week vendor study.

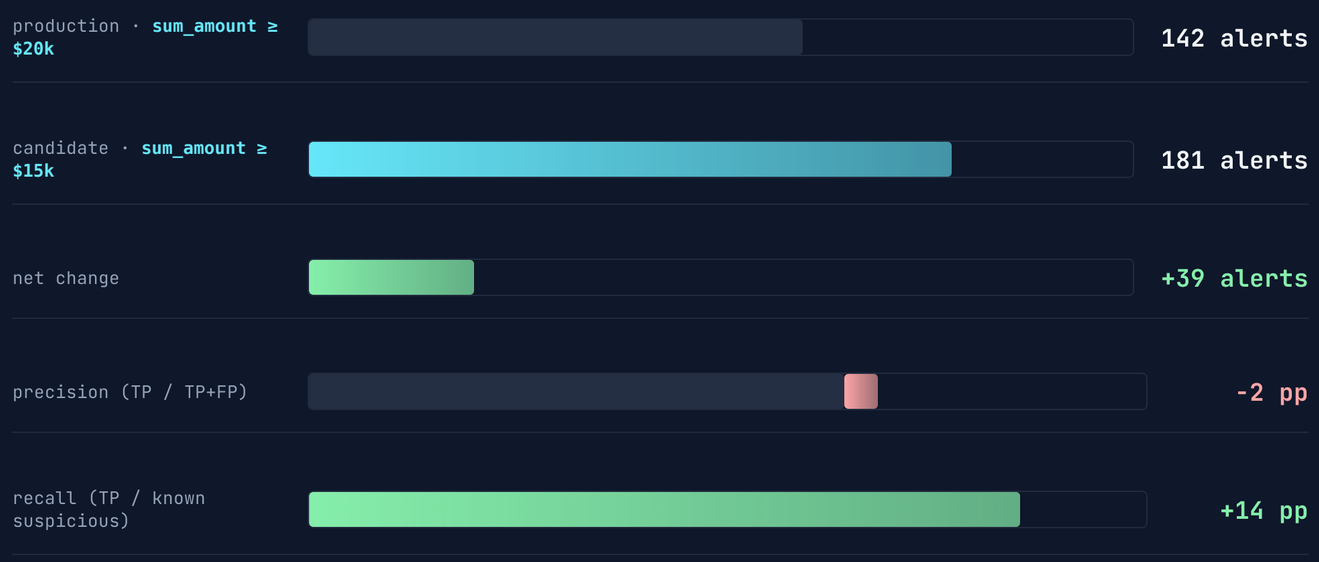
01
Backtester replays last quarter's data against a candidate threshold change. Returns precision / recall / alert delta — the exact evidence SR 26-2 and OSFI E-23 ask for.

02
Tune the spec, not the SQL. The tuning grid lives in the rule itself, alongside the regulation citation. Grid sweep is one command; outputs a YAML patch you can hand to 1LoD.

03
Then propose-change. The delta becomes the rationale paragraph in the 2LoD packet. The packet attaches to the spec PR. *Same loop, every threshold, every quarter.*

Model challenge stops being a vendor engagement. It becomes a Tuesday afternoon.

AML BACKTEST · STRUCTURING_CASH_DEPOSITS · 4 QUARTERS



ILLUSTRATIVE · YOUR NUMBERS DEPEND ON YOUR DATA · aml backtest --rule X --quarters 4

NET NEW STRS / MO	ANALYST TIME	SR 26-2 EVIDENCE
+1	+2.3h	✓

ACT IV · THE DETECTION SUBSTRATE

When the auditor asks "who decided?" the answer is in the file.

01 · SPEC

COMPLIANCE COMMITS

aml.yaml signed by source-control

`sha256: a73b8e2f...4cd9`

SHA-256 of the exact YAML at commit time. Pinned in every audit ledger.

→

02 · RUN

ENGINE PRODUCES

manifest.json + decisions.jsonl

`sha256: 0f2a17c9...ea84`

Each decision (alert fired, case routed, STR drafted) gets its own record + previous-line hash.

→

03 · VERIFY

AUDITOR / 2LOD

Replay byte-for-byte

```
aml replay run-2026-04-29
```

Re-run the same spec on the same data with the same seed; outputs must hash-match.

✓

VERIFIED RUN · 2026-04-29

Hash chain integrity verified — `1,847` decisions re-hashed, all match.

✗

IF A SINGLE LINE IS ALTERED

`HashChainBroken at line 412` — every downstream attestation refuses to build.

"Cannot evidence" *is the most common enforcement finding. Hash-chained ledgers + replay are the answer in code, not a policy.*

ACT V · INSTITUTION-FIT

Zero to day-one detection — before the kettle boils.

01 **Pick an archetype.** Schedule-I bank · community bank · VASP · fintech. Each scaffolds with sensible contracts, three starter rules, four queues, and a STR form pre-cited.

02 **Map your warehouse.** `aml byod` inspects your CSV/Parquet/DuckDB schemas and proposes the column mapping with confidence scores. Low-confidence rows are flagged, not auto-decided.

03 **Validate, then run.** `aml validate-data` checks completeness against your spec — every column, every quality check, before the engine touches a row.

Most pilots stall on "where do we even start?". This is the path from fresh repo to a regulator-mappable spec running on real data — same afternoon.

aml init · aml byod · aml validate-data

```
$ aml init /tmp/acme --archetype schedule_i_bank \
   --jurisdiction CA --non-interactive

✓ Scaffold written to /tmp/acme
  → /tmp/acme/aml.yaml
  → /tmp/acme/README.md

$ aml byod /tmp/acme/aml.yaml ./warehouse-export/

Mapping report · aml.yaml
```

Contract	Source	Total	Mapped	LowConf	Unmapped
txn	txn.csv	7	7	0	0
customer	customer.csv	5	5	0	0

```
Mapping written data_mapping.yaml
```

ACT V · INSTITUTION-FIT

Escalations stop drowning in unread email.

01 **Cron'd into your channel.** Every 4-24h, a rendered digest of escalated cases, breached SLAs, and newly-aging queues drops into Slack or Teams.

02 **Suppressed when nothing fires.** `--suppress-empty` skips the post when the window is clean — channels stay high-signal.

03 **Webhook-only.** No vendor connector, no SaaS forwarder. Slack/Teams incoming-webhook URLs in env vars, one egress destination, your perimeter.

The supervisor finds out about a breach when it happens, not at end-of-day. That alone is most of the "why we keep adding tools" problem solved.

aml notify-digest · dry-run · canadian_schedule_i_bank

```
$ aml notify-digest spec.yaml \
   --since-hours 24 --dry-run

──────────────── Digest preview ────────────────

*AML Digest · schedule_i_bank_aml*
  (24h ending 2026-05-02T23:44)

Total events: *1*
· unknown: 1

· *escalated*
  rule:     dormant_account_activity
  customer: ?
  case:     dormant_account_activity
              C0007 · 2026-04-24T160811
```

ACT V · INSTITUTION-FIT

Same engine. Four regulators.

UNITED STATES · US

FinCEN · BSA · 31 CFR 1010

SAR via FinCEN E-File · CTR · BOI rule (Mar 2025 narrowing tracked by `aml regwatch`)

examples/community_bank · examples/us_rtp_fednow

CANADA · CA

FINTRAC · OSFI · PCMLTFA / PCMLTFR

STR · LCTR · EFTR · OSFI E-23 model risk · `aml export-amla-str` & goAML

examples/canadian_schedule_i_bank

UNITED KINGDOM · GB

FCA · POCA · PSR PS24/3

SAR via NCA · APP fraud reimbursement (Oct 2024) · `app_fraud_first_use_payee` typology

examples/uk_bank · examples/uk_app_fraud

EUROPEAN UNION · EU

EBA · AMLA · AMLD6 · AMLR

AMLA RTS draft 2026-02 outcomes pack · goAML XML · cross-border sandbox per FATF R.18

examples/eu_bank

FATF · INTERNATIONAL

R.16 Travel Rule · R.18 Sharing

Notabene + Sumsub adapters · cross-border typology sandbox · ISO 20022 ingest

src/aml_framework/integrations/

DIGITAL ASSETS · VASP

VASP · MiCA · NYDFS · MAS

Blockchain heuristics, on/off-ramp risk, mixer/sanctioned-cluster detection

examples/crypto_vasp

FRAUD · AML CONVERGENCE

Cyber-enabled fraud · FATF 2026

Pig-butchering, mule-funnel, RTP/FedNow push fraud — fraud-AML case linkage in one queue

examples/cyber_enabled_fraud

TRADE-BASED ML

FATF R.32 · WCO · BIS

Invoice mismatch, duplicate documents, trade-finance round-tripping

examples/trade_based_ml

9th SPEC SLOT · 2026 H2

Australia · AUSTRAC

Tier-2 designated services scope expansion, Q3 2026 — pre-flighted with `aml init --jurisdiction AU`

examples/au_designated_services · planned

Pick a jurisdiction. Same spec shape, same CLI, same audit chain.

ACT VI · THE TECH SUBSTRATE

One spec compiles into everything else.

01 · INPUT

Spec + Data

`aml.yaml + warehouse`

JSON Schema (structural) + Pydantic v2 cross-reference checks. CSV/Parquet/DuckDB/S3/GCS/Snowflake/BigQuery sources. ISO 20022 ingest.

`spec/`
`data/sources.py`

02 · ENGINE

Run

`DuckDB · seeded · deterministic`

Same spec + same data + same seed = byte-identical output hashes. 4 rule logic types: aggregation_window, custom_sql, python_ref, list_match.

`engine/runner.py`
`engine/audit.py`

03 · CASES

Investigation

`aggregator · SLA · linkage`

Three aggregation strategies. Fraud↔AML case linkage. STR bundling. SLA timer per queue. Persona-aware routing.

`cases/`
`cases/sla.py`

04 · GENERATORS

Artefacts

`SQL · DAG · audit · STR`

Audit pack (jurisdiction-mapped) · MRM bundle (SR 26-2) · effectiveness pack (FinCEN NPRM) · STR narratives · DAG stubs · control matrix.

`generators/`
`narratives/`

05 · SURFACES

Read

`dashboard · API · CLI`

Streamlit dashboard (26 pages, persona-aware). FastAPI REST with JWT/OIDC + multi-tenant. Typer CLI: 24 commands, 1,632 tests.

`dashboard/`
`api/ · cli.py`

CROSS-CUTTING | `compliance/regwatch` reg-text drift · `integrations/notabene + sumsub` Travel Rule · `compliance/sandbox` FATF R.18 · `pkyc/` periodic KYC · `vasp/` blockchain heuristics

ACT VI · THE TECH SUBSTRATE

Reproducible by command. Tested by CI.

BUILD · RUN · VALIDATE

`aml init`	Scaffold a starter spec by archetype.
`aml byod`	Map your warehouse into spec contracts.
`aml validate`	JSON Schema + Pydantic + cross-ref check.
`aml validate-data`	Quality checks against your spec.
`aml run`	Execute detectors · seeded · deterministic.
`aml replay`	Verify a historical run byte-for-byte.

DETECTION · TUNING

`aml typology-list`	Browse the curated catalogue (8 + growing).
`aml typology-import`	Atomic install of a regulator-cited rule.
`aml add-rule`	Wizard for structuring / velocity / juris rules.
`aml backtest`	4-quarter precision/recall on a candidate.
`aml tune`	Sweep tuning grid; emit YAML patch.
`aml propose-change`	2LoD packet with diff + sign-off block.

OPERATIONS · DAILY

`aml today`	Per-persona morning checklist.
`aml dashboard`	Streamlit · 26 pages · persona-aware.
`aml notify-digest`	Slack/Teams escalation digest.
`aml regwatch`	Watch reg pages for silent narrowing.
`aml sanctions-sync`	Refresh OFAC / UN / EU lists.
`aml pkyc-scan`	Periodic KYC trigger sweep.

AUDIT · REGULATOR SURFACES

`aml auditor-pack`	Self-service ZIP · hash-verified.
`aml audit-pack`	Examination ZIP · per jurisdiction.
`aml mrm-bundle`	SR 26-2 model dossier per rule.
`aml effectiveness-pack`	FinCEN NPRM 2026 evidence.
`aml export-amla-str`	AMLA RTS draft 2026-02 · JSON.
`aml export-goaml`	goAML XML · UNODC reporting.

1,632
TESTS · ALL GREEN

5
CI JOBS · PER PR

ACT VII · CLOSE

What's actually in the box.

CLI COMMANDS

24

One answer for every leader question — board, regulator, analyst.

CURATED TYPOLOGIES

8

All cited · TD 2024 · FATF 2026 · UK PSR · FinCEN.

EXAMPLE SPECS

9

US · CA · UK · EU · VASP · Trade ML · Cyber-fraud · RTP/FedNow.

DASHBOARD PAGES

26

Persona-aware · sidebar filters by role.

TESTS · ALL GREEN

1,632

5 CI jobs · lint · unit · api · e2e · docker.

ACTIVE JURISDICTIONS

4 *+1*

US · CA · UK · EU + FATF Travel Rule · AU planned.

AUDIT CHAIN

SHA *-256*

Hash-chained `decisions.jsonl` · replayable byte-for-byte.

LICENCE

MIT

Runs in your perimeter · no per-seat fees · forkable.

No vendor lock-in. **No black-box rules.** *Every line defensible to a regulator who walks in tomorrow.*

END · OR BEGINNING

Three paths *from here.*

READ · 15 MIN

"Show me what's there."

For CCO/MLRO doing competitive scan, or 2LoD doing tooling diligence.

→ docs/research/2026-04-aml-process-pain.md
→ docs/case-studies/td-2024.md
→ tomqwu.github.io/aml_open_framework_demo

TRY · 5 MIN

"Run it on my desk."

For engineers, data folks, or curious analysts. No vendor demo cycle.

→ git clone · pip install -e ".[dev,dashboard,api]"
→ aml demo --persona cco
→ aml dashboard examples/community_bank/aml.yaml

TALK · 30 MIN

"Walk me through it."

For pilots, evaluations, or anyone wanting a tour of a specific surface.

→ open a Discussion · share your spec
→ tag @tomqwu in an issue
→ pull-request a typology you'd like in the catalogue

github.com/tomqwu/aml_open_framework